

SVAN M2 QUADRUPEd ROBOT

MTP Running Report
Dynamic Modeling and Parameter Analysis

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1 Short summary of the previous work

The Svan M2 has four legs and three actuated joints in each leg:

$$\text{HAA} \rightarrow \text{HFE} \rightarrow \text{KFE}.$$

Therefore the robot has 12 actuated joints. With the floating body included, the generalized coordinates have 18 DOF:

$$6 \text{ body DOF} + 12 \text{ joint DOF} = 18.$$

The URDF model was used in MuJoCo and the FK/IK results were checked against the simulation. We also tested a basic standing PD controller and a trot gait.

2 Parameters of the Svan M2

The main system-level specifications from the xTerra Svan M2 User Manual are:

Parameter	Value
Total mass (manual)	12 kg
Actuators	12 $\times$ QDD A2
Maximum torque, HAA/HFE	11.92 N·m
Maximum torque, KFE	23.84 N·m
Maximum dimensions	46.8 $\times$ 43.6 $\times$ 46.2 cm
Maximum linear speed	0.7 m/s
Maximum payload	5 kg
Operating voltage	24 V
Onboard computer	Raspberry Pi 4B

For the dynamics, the important parameters are not only the total mass. For each rigid link we need

$$m_i, r_{Ci}, I_i,$$

where m_i is the link mass, r_{Ci} is the centre-of-mass location in the link frame and I_i is the inertia tensor about the CoM.

These values are present in the generated Svan M2 URDF. For example, the base link contains

$$m_B = 5.18596663 \text{ kg},$$

with CoM offset

$$r_{CB} = [0.00918208 \quad -0.00133606 \quad -0.00430117]^T \text{ m}.$$

For one leg, the extracted link masses are:

Link	Mass (kg)
Hip	1.12704463
Thigh	0.15396568
Shank	0.12448200
Foot	0.02330447

The four legs use the corresponding symmetric parameter sets.

2.1 Mass check

Using the masses in the generated URDF,

$$M_{URDF} = 5.18596663 + 4(1.12704463 + 0.15396568 + 0.12448200 + 0.02330447) = 10.90115375 \text{ kg.}$$

The manual gives 12 kg. Therefore,

$$\Delta M = 10.90115375 - 12 = -1.09884625 \text{ kg}, \quad \frac{\Delta M}{12} \times 100 = -9.157\%.$$

This discrepancy is now recorded as a parameter-validation point. At present we use the URDF values as the simulation parameters, but we should confirm the difference with the lab/xTerra before using the model for hardware experiments.

3 Single-leg kinematic model

For one leg, let

$$q_l = [q_1 \quad q_2 \quad q_3]^T$$

where q_1 is HAA, q_2 is HFE and q_3 is KFE.

The joint axes at the URDF zero configuration are

$$a_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad a_2 = a_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

The foot position can be written using the successive rotations and URDF link offsets as

$$p_F = p_{HAA} + R_x(q_1)d_1 + R_x(q_1)R_y(q_2)d_2 + R_x(q_1)R_y(q_2)R_y(q_3)d_3.$$

Since both HFE and KFE use the same pitch axis,

$$R_y(q_2)R_y(q_3) = R_y(q_2 + q_3).$$

Hence, defining $q_{23} = q_2 + q_3$,

$$p_F = p_{HAA} + R_x(q_1)d_1 + R_x(q_1)R_y(q_2)d_2 + R_x(q_1)R_y(q_{23})d_3.$$

This is the kinematic relation used for the leg model.

4 Dynamic model of one leg

The dynamics were derived using the Lagrangian method,

$$\mathcal{L} = T - V.$$

For link i ,

$$\dot{p}_{Ci} = J_{vi}(q)\dot{q}, \quad \omega_i = J_{\omega i}(q)\dot{q}.$$

The kinetic energy of a link is

$$T_i = \frac{1}{2}m_i\dot{p}_{Ci}^T\dot{p}_{Ci} + \frac{1}{2}\omega_i^T I_i^W \omega_i,$$

where

$$I_i^W = R_i I_i^B R_i^T.$$

Therefore,

$$T = \frac{1}{2} \dot{q}^T M(q) \dot{q},$$

with

$$M(q) = \sum_i (m_i J_{vi}^T J_{vi} + J_{\omega i}^T I_i^W J_{\omega i}).$$

For gravity,

$$V = \sum_i m_i g z_{Ci}, \quad \text{so} \quad G(q) = \frac{\partial V}{\partial q}.$$

The Coriolis/centrifugal term was obtained from the same $M(q)$ using the Christoffel symbols

$$\Gamma_{ijk} = \frac{1}{2} \left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i} \right),$$

so that

$$[C(q, \dot{q})\dot{q}]_i = \sum_{j=1}^n \sum_{k=1}^n \Gamma_{ijk} \dot{q}_j \dot{q}_k.$$

Thus the fixed-base leg equation is

$$M_l(q_l) \ddot{q}_l + C_l(q_l, \dot{q}_l) \dot{q}_l + G_l(q_l) = \tau_l.$$

5 Explicit 3-DOF leg result

Let $\varphi = q_2 + q_3$. The derived leg mass matrix is

$$M_l(q) = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{12} & M_{22} & M_{23} \\ M_{13} & M_{23} & M_{33} \end{bmatrix}.$$

Some of the entries are

$$\begin{aligned} M_{12} &= m_2 L_1 l_{c2} \sin q_2 + m_3 (L_1 L_2 \sin q_2 + L_1 l_{c3} \sin \varphi), \\ M_{13} &= m_3 L_1 l_{c3} \sin \varphi, \\ M_{22} &= I_{2,yy} + I_{3,yy} + m_2 l_{c2}^2 + m_3 (L_2^2 + 2L_2 l_{c3} \cos q_3 + l_{c3}^2), \\ M_{23} &= I_{3,yy} + m_3 l_{c3} (L_2 \cos q_3 + l_{c3}), \\ M_{33} &= I_{3,yy} + m_3 l_{c3}^2. \end{aligned}$$

The remaining M_{11} terms were obtained from the same kinetic-energy expression and include the abduction-axis inertia and the configuration-dependent inertial terms.

The gravity vector is $G(q) = [G_1 \ G_2 \ G_3]^T$, with

$$\begin{aligned} G_1 &= g \left[m_1 l_{c1} \cos q_1 + m_2 (L_1 \cos q_1 + l_{c2} \sin q_1 \cos q_2) + m_3 (L_1 \cos q_1 + L_2 \sin q_1 \cos q_2 + l_{c3} \sin q_1 \cos \varphi) \right], \\ G_2 &= g \cos q_1 \left[m_2 l_{c2} \sin q_2 + m_3 (L_2 \sin q_2 + l_{c3} \sin \varphi) \right], \\ G_3 &= g m_3 l_{c3} \cos q_1 \sin \varphi. \end{aligned}$$

The symbolic result was checked for symmetry: $M(q) = M(q)^T$. The model also reduces to the standard 2-link planar case when the HAA motion is removed and the corresponding out-of-plane inertial terms are neglected.

6 Numerical check using the URDF parameters

Using the extracted parameters, the total URDF mass is $M_{URDF} = 10.90115375$ kg. For the FR leg at the zero configuration, the fixed-base numerical mass matrix used for checking the symbolic result was

$$M_{FR}(0) = \begin{bmatrix} 0.023689920 & -0.000022131 & -0.000022131 \\ -0.000022131 & 0.012455245 & 0.003592358 \\ -0.000022131 & 0.003592358 & 0.001652622 \end{bmatrix}.$$

The corresponding gravity vector was

$$G_{FR}(0) = \begin{bmatrix} -1.091133 \\ 0.002681 \\ 0.002681 \end{bmatrix} \text{ N} \cdot \text{m}.$$

These values are mainly used as a numerical consistency check between the symbolic derivation and the URDF-based model. Small nonzero residuals in M_{12} , M_{13} , G_2 and G_3 at the zero configuration come from off-axis CoM components in the actual URDF that are not captured by the simplified single-axis symbolic model.

In this numerical calculation, the foot mass and inertia were included explicitly as a separate fixed terminal link. Therefore, the 0.02330447 kg foot mass was not absorbed into the shank parameters.

7 Extension to the complete quadruped

For the complete robot,

$$q = \begin{bmatrix} q_B \\ q_j \end{bmatrix}, \quad q_B \in \mathbb{R}^6, \quad q_j \in \mathbb{R}^{12}.$$

The complete mass matrix is partitioned as

$$M(q) = \begin{bmatrix} M_{BB} & M_{Bj} \\ M_{jB} & M_{jj} \end{bmatrix},$$

with $M_{BB} \in \mathbb{R}^{6 \times 6}$, $M_{Bj} \in \mathbb{R}^{6 \times 12}$ and $M_{jj} \in \mathbb{R}^{12 \times 12}$.

Without foot contact,

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = S^T \tau$$

where $S = \begin{bmatrix} 0_{12 \times 6} & I_{12} \end{bmatrix}$.

When feet are in contact with the ground,

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) + J_c(q)^T \lambda = S^T \tau.$$

For a stationary contact,

$$J_c(q)\dot{q} = 0,$$

and therefore

$$J_c(q)\ddot{q} + \dot{J}_c(q, \dot{q})\dot{q} = 0.$$

We are not expanding an 18×18 symbolic matrix by hand. For the complete robot, the practical method is to use the URDF with a rigid-body dynamics implementation based on CRBA/RNEA [6, 7] and then compare the numerical result against simulation.

8 Present status and next steps

Completed so far

1. Svan M2 URDF model tested in MuJoCo.
2. FK and IK verified against simulation.
3. Preliminary standing PD controller and basic trot gait tested.
4. Link masses, CoM locations and inertia tensors identified from the generated URDF.
5. URDF total mass checked against the 12 kg manual value; a 9.157% difference was found.
6. Single-leg Lagrangian dynamic model derived, including $M(q)$ and $G(q)$ and the construction of $C(q, \dot{q})\dot{q}$.
7. Full floating-base equation with contact forces written for the 18-DOF robot.

Next work

1. Confirm the mass discrepancy with the SIL lab/xTerra and freeze the parameter set used for simulation.
2. Run the complete numerical 18-DOF dynamics from the final URDF and compare with MuJoCo.
3. Start the baseline PID/path-following controller using the dynamics model for gravity/feedforward compensation.
4. Continue with ROS 2/Gazebo integration after the dynamics and controller are checked.

Sources used:

1. xTerra Robotics, Svan M2 User Manual, V2026.1.2.
2. xTerra Robotics, official m2_sdk repository, generated Svan M2 URDF (svan_metal.urdf).

References

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